

EUROPEAN CAR RECOMMENDATION SYSTEM

Cloud Computing Final Project

TEAM MEMBERS

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aws



Project Completion Status

- Built a Streamlit web application
- Hosted the application on AWS EC2
- Connected AWS RDS MySQL database
- Configured Security Groups
- Uploaded source code to GitHub

Future Improvements

- **HTTPS**
- **Auto Scaling**
- **Load Balancer**

Requirements Covered

Functional

- Search and filter cars
- Display recommended cars

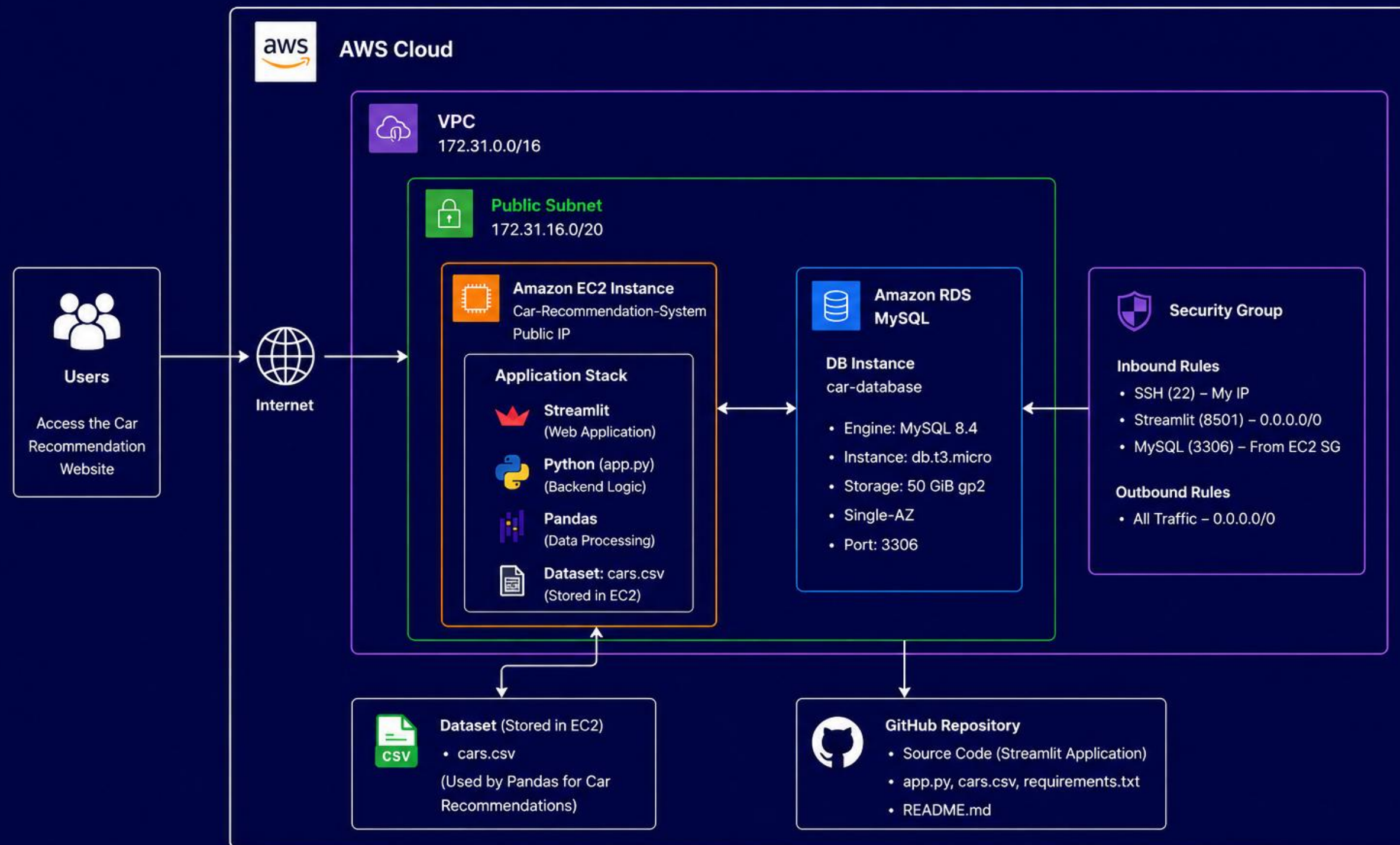
Non-functional

- Public cloud deployment
- Managed database storage

Architecture Diagram

European Car Recommendation System – Architecture

Deployed on AWS EC2 using Streamlit and AWS RDS MySQL



AWS Services Used

- Amazon EC2
- Amazon RDS
- Security Groups
- GitHub

Budget Estimation

Estimated monthly cost for running the European Car Recommendation System on AWS

Service	Monthly Cost (USD)
 EC2	~\$15
 RDS (MySQL)	~\$12
 Storage (EBS / S3)	~\$5
 Total	~\$34



Cost Breakdown

- **EC2 (~\$15)**
t2.micro instance running the Streamlit application.
- **RDS (~\$12)**
MySQL database instance for storing car data.
- **Storage (~\$5)**
EBS volume for EC2 and additional storage.



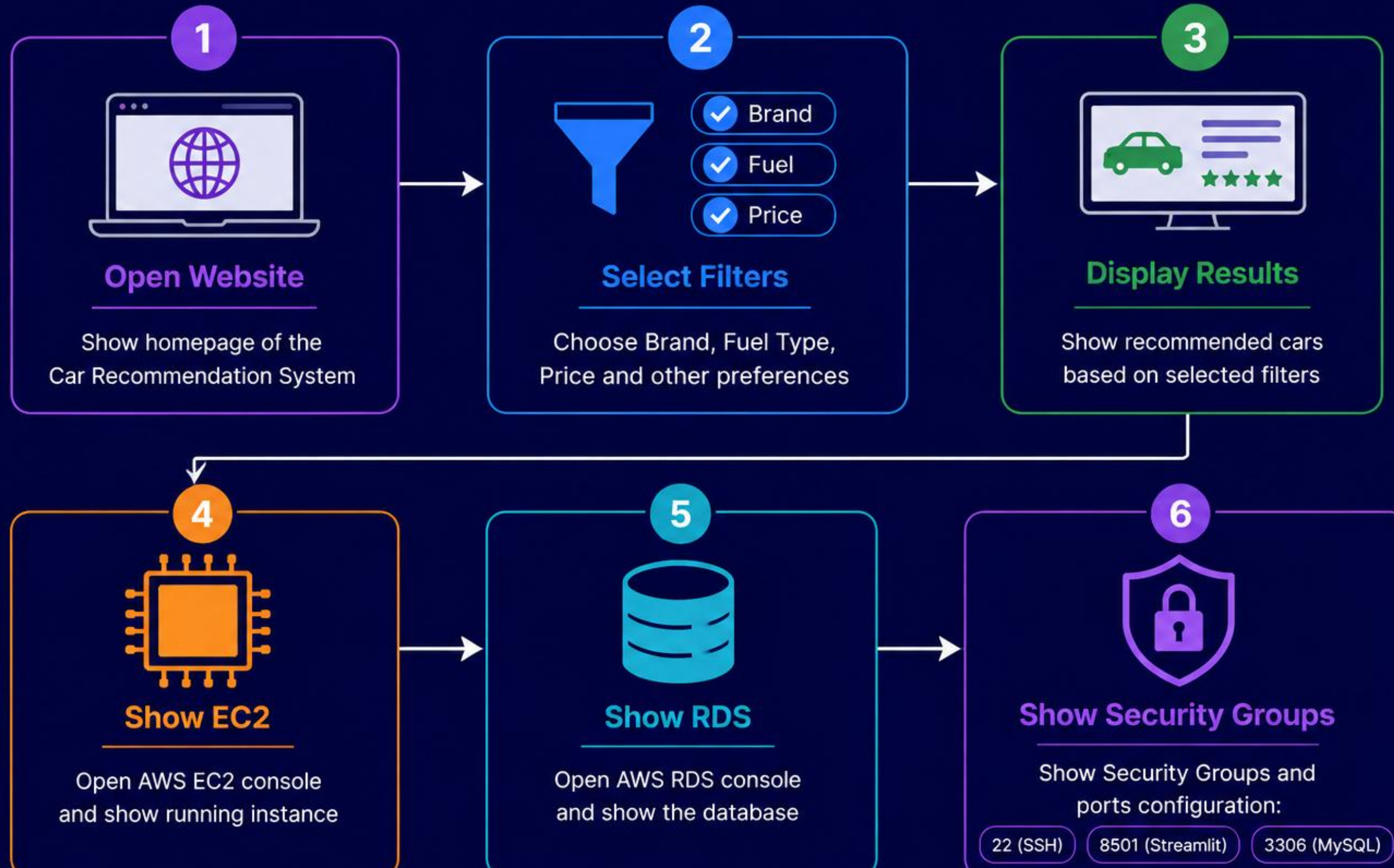
Total estimated monthly cost: ~\$34–\$35

This cost may vary slightly based on usage and data transfer.



Live Demonstration Flow

Step-by-step walkthrough of the running system



Individual Contribution

This slide highlights the specific technical work and contributions made by this team member towards building the European Car Recommendation System.

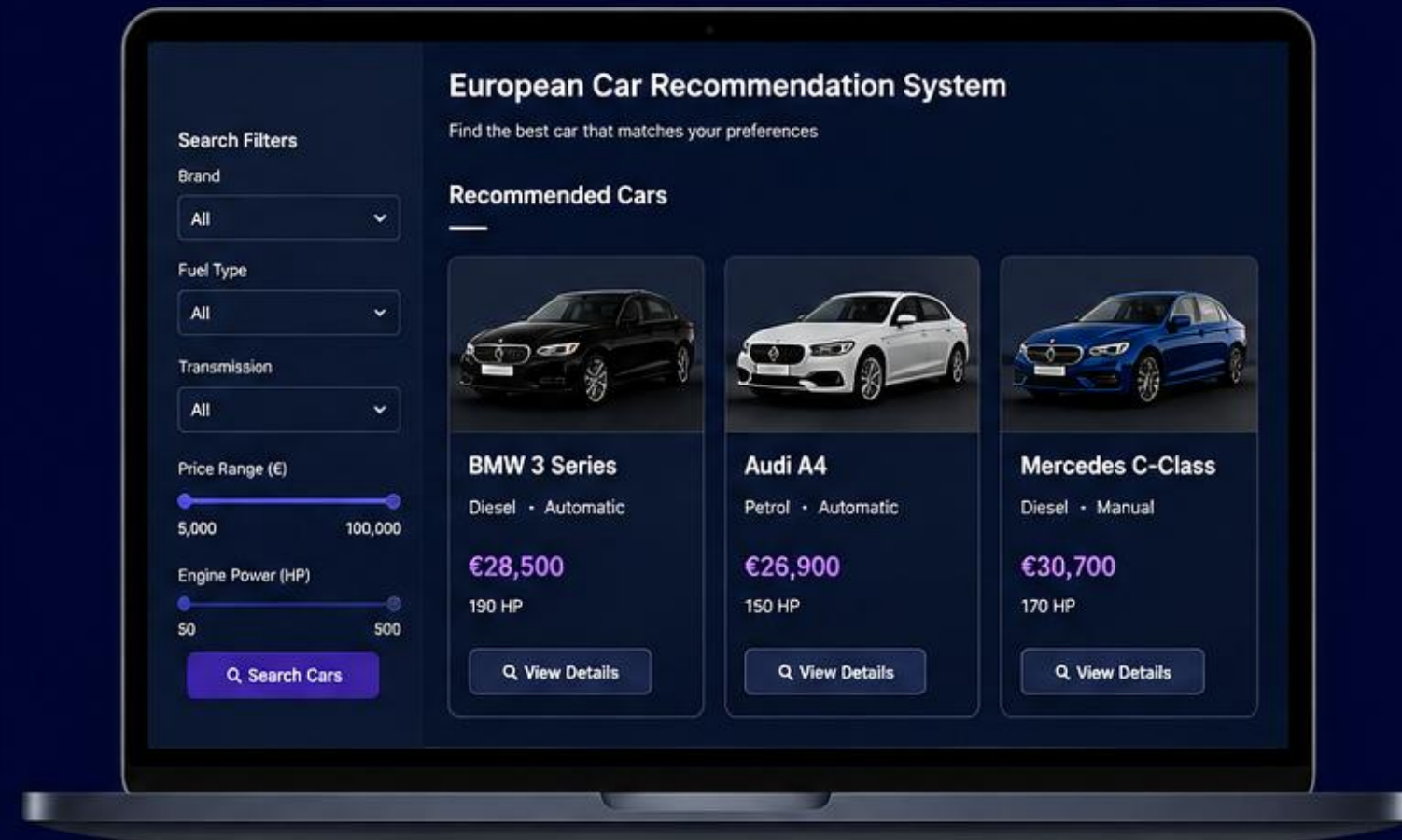


Frontend Development & User Interface

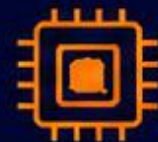


Technical Implementation

- ✓ Developed the Streamlit web application.
- ✓ Designed the user-friendly interface and page layout.
- ✓ Implemented car search and filtering options.
- ✓ Displayed recommended cars based on user inputs.
- ✓ Improved the overall user experience and application design.



User Access
via Browser



Streamlit App
on EC2



Filter Preferences
(Brand, Price, etc.)



Fetch Data
from RDS MySQL



Show Recommended
Cars to User

Individual Contribution

This slide highlights the specific technical work and contributions made by this team member towards building the European Car Recommendation System.



AWS EC2 Deployment & Configuration

Technical Implementation

- ✓ Created and configured the Amazon EC2 t3.small Ubuntu instance.
- ✓ Installed Python, Streamlit, and all required libraries and dependencies.
- ✓ Deployed the Streamlit application on the EC2 server.
- ✓ Configured SSH access and security groups to allow required ports (22, 8501).
- ✓ Verified that the website is accessible online and running successfully.



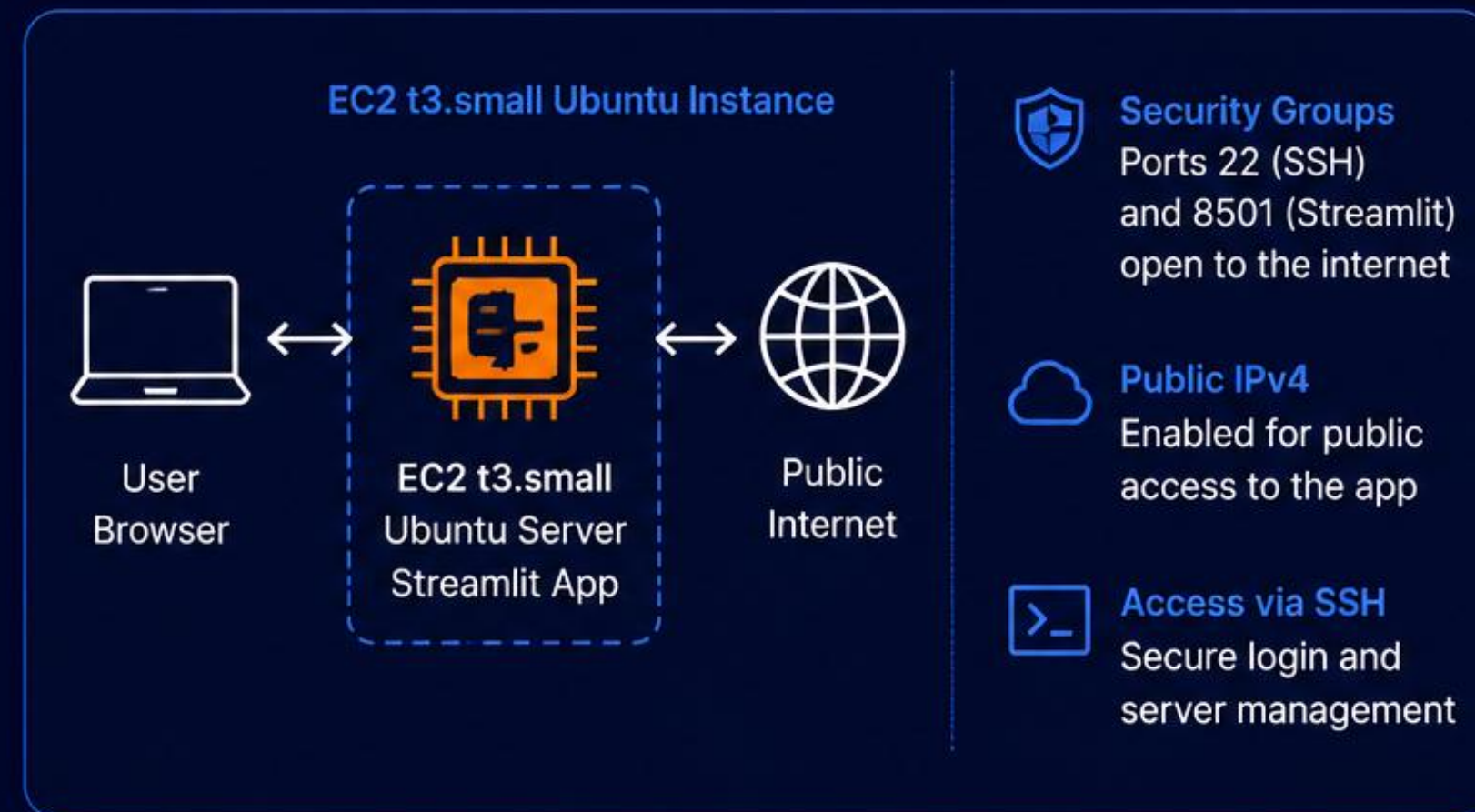
EC2



Streamlit



Python



User Access
via Browser



Application Hosted
on EC2 (Streamlit)



Process User Requests
on EC2



Fetch Data from
RDS MySQL



Response Sent Back
to User

Individual Contribution

This slide highlights the specific technical work and contributions made by this team member towards building the European Car Recommendation System.



AWS RDS Database & Backend Logic

Technical Implementation

- ✓ Created and configured the AWS RDS MySQL database instance.
- ✓ Designed the database schema and uploaded the car dataset.
- ✓ Connected the Streamlit application to the RDS MySQL database.
- ✓ Implemented backend logic to fetch and filter car data based on user inputs.
- ✓ Tested database queries and ensured successful connectivity and performance.



RDS MySQL Architecture



Streamlit Application



Amazon RDS MySQL

db.t3.micro
50 GiB gp2 Storage



Car Data Tables



Managed Database
AWS RDS MySQL for secure and reliable data storage.



Secure Access
Database access restricted to EC2 security group.



Automated Backups
Daily backups with point-in-time recovery enabled.



High Availability
Single-AZ deployment for consistency and uptime.



User Inputs
(Filters & Preferences)



Streamlit App
(Frontend)



Python Backend
(Query Processing)



RDS MySQL
(Data Retrieval)



Filtered Car Data
(Based on Inputs)



Recommended Cars
Displayed to User

Individual Contribution

This slide highlights the specific technical work and contributions made by this team member towards building the European Car Recommendation System.



Testing, Documentation & GitHub Management

Technical Implementation

- ✓ Tested the complete application workflow from frontend to database.
- ✓ Verified EC2, RDS connectivity and network configurations.
- ✓ Managed the project repository on GitHub and maintained code versions.
- ✓ Prepared project documentation including architecture, configurations and cost.
- ✓ Performed debugging, issue resolution and final system validation.



GitHub



Testing



Documentation

What I Delivered



Comprehensive Testing

Tested all features, filters and recommendations to ensure accuracy and reliability.



System Validation & Security

Validated security groups, access controls and overall system security.



GitHub Management

Organized project files, maintained commits and ensured code integrity.



Documentation

Created architecture diagrams, setup guides and user instructions.



Final Validation

Performed end-to-end validation and ensured the system is ready for demo.



Test All Components



Validate Security & Connectivity



Push Code to GitHub



Prepare Documentation



Final System Validation



Ready for Demo

Final Project Summary



PROJECT OVERVIEW

- Developed a European Car Recommendation System using Streamlit.
- Deployed the application on AWS EC2.
- Stored car data in AWS RDS MySQL.



AWS SERVICES USED

-  Amazon EC2
-  Amazon RDS (MySQL)
-  Security Groups
-  GitHub Repository



PROJECT OUTCOME

- Successfully deployed a cloud-hosted web application.
- Users can filter cars and receive recommendations.
- Verified connectivity between EC2 and RDS.



KEY LEARNINGS

- Learned cloud application deployment using AWS.
- Gained experience with EC2, RDS, networking, and Security Groups.
- Understood the complete deployment workflow from development to cloud hosting.



Successfully developed and deployed a cloud-based **European Car Recommendation System** using AWS services.



Users



Internet



Amazon EC2
(Streamlit App)



Amazon RDS
(MySQL)

Thank You!

We appreciate your time and attention.



Questions?

We are happy to answer!



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Built on AWS



Reliable
Secure & Scalable



Efficient
Optimized Performance



Secure
Safe & Protected

Thank You!

For your support!